



Homepage Search Component Data Flow

Complete architecture analysis of the Bhoomitali search system

🎨 UI Layer - User Interface Components

SearchContainer.tsx

Main search orchestrator with gradient header background. Integrates all search sub-components and handles view details/contact owner actions.

SearchHeader.tsx

Primary search input with location selection. Features property code detection (6-char alphanumeric), real-time suggestions, and responsive design.

SearchFilters.tsx

Property type, BHK, and price range filters with dropdown interfaces for refined search criteria.

PropertyMapHomeView.tsx

Homepage integration point containing search interface and map/listing panels with responsive split layout.



⚡ Logic Layer - Hooks & State Management

useSearch.ts

Central search state management with smart search detection (6-char property codes), filter management, and auto-search triggers when filters are cleared.

useSearchFilters.ts

Filter state management handling search query, location, and filter updates with backend compatibility transformations.

usePropertyMapData.ts

Homepage-specific data management for property listings and map integration with automatic loading of latest properties on mount.



Service Layer - Business Logic & API

searchService.ts

Primary search service with property code validation, database integration, and performance monitoring. Handles smart search strategy selection.

realSearchService.ts

Real search implementation with direct Supabase integration for live property data retrieval and filtering.

searchDataAdapter.ts

Data transformation layer converting database results to frontend format with property type mapping and result normalization.

searchFallbackService.ts

Fallback service ensuring search functionality continues even when primary services encounter issues.



Database Layer - Supabase Functions & Queries

search_residential_properties()

Searches residential properties with comprehensive filtering (price, bedrooms, bathrooms, area) and pagination support.

search_commercial_properties()

Handles commercial property searches including rent, sale, and coworking spaces with area and price filtering.

search_land_properties()

Specialized land property search with land type, area, and location-based filtering capabilities.

search_property_by_code()

Direct property lookup using 6-character alphanumeric codes with case-insensitive matching.

get_latest_properties()

Loads default 50 properties for homepage display when no search filters are active.

CodeSearchDb.ts

Database adapter for property code searches with validation and error handling.



Key Search Features & Flow Logic

Smart Search Detection: Automatically detects 6-character alphanumeric property codes and uses dedicated search strategy

Real-time Suggestions: 300ms debounced search suggestions with code-aware recommendations (max 5 results)

Multi-Property Type Support: Handles residential, commercial, and land properties with flow-specific filtering

Auto-Clear Behavior: Automatically loads latest properties when filters are cleared

Responsive Design: Mobile: stacked layout | Desktop: horizontal layout with location, search bar, and button

Performance Optimized: Built-in search performance tracking and efficient database queries with pagination

Map Integration: Synchronizes search results with map display and property markers

Visual Feedback: Property codes get special styling (orange border, "CODE" badge)